

KITEWRIGHT RANGE · DESIGN DECISIONS REGISTER

Kitewright Range design decisions register

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DOCUMENT CONTROL

Document ID	KWR-DEC-001	Version	0.3
Status	Draft	Date	2026-10-04
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kitewright-range		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-04	Amish Chadha	Round-3 decision 10A (Amish, 2026-10-04: "For round 3, I agree with all your proposed recommendations") recorded in Decisions made (KWR-DDR-004)
0.2	2026-10-03	Amish Chadha	01 to 04 decided by Amish on 2026-10-03 ("i agree with all the 46 recommendations you provided. please proceed."; 36B, 37A, 38A) and moved to Decisions made (KWR-DDR-003); Core power leads taken on under Core decision 33B
0.1	2026-10-03	Amish Chadha	Register opened at TRL 3; decisions made under the 2026-10-03 pre-approval; four requirement decisions proposed, awaiting Amish

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Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`, and the open ones in `docs/REVIEW.md` (TRL 3, Decisions for Amish); this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. Amish decided O1 to O4 on 2026-10-03 (KWR-DDR-003) and the Kitewright family reconciliation on 2026-10-04 (KWR-DDR-004); the design and build plan carry those decisions. R2 (30.6 min) and R7 (11.35 kg) remain not met, as accepted for the first prototype. Nothing in carrying out 10A needs Amish.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	5215-class lift motor thrust with a 22 x 7 in propeller at 0.736 kg/m ³ , measured in AltiRig	R1 rests on maker-class figures scaled by density; the 52.0 % margin allows about 14 % less thrust	KWR-CAL-001 C, KWR-DDR-003
2	Cold ratings of the ESCs, servos, cruise motor ESC and pitot heater to -20 °C, or a cold soak test	R5 is met only by specification	KWR-REQ-001
3	The joiner slides into both spar bores without play	The wing joint depends on it	KWR-DDR-002, C6
4	The cruise motor's bolt circle and shaft length	They set the nose cone inserts and the propeller position	KWR-DDR-002, C5
5	Kitewright Core avionics envelope (180 x 110 x 60 mm) and mass (0.78 kg with the mast); Core's own figure after its decision 33B is about 0.99 kg including the payload mount (Range carries 0.90 kg for the same items)	They set the front bay and the balance	Cross-repo assumption; <code>docs/REVIEW.md</code> cross-repo actions
6	ColdCell pack size (138 x 75 x 82 mm), mass (1.40 kg) with 5.0 Ah high-rate 21700 cells, and its lead (XT90 here, AS150 in the Core)	They set the battery bays, the energy and the balance	Cross-repo assumption, KWR-DDR-003
9	Kitewright Core pad positions and grommets for the four 8 AWG power leads Range now supplies	Lead lengths and the solder joints at integration	Core decision 33B, KWR-DDR-003
7	Payload mount plate (160 x 80 x 12 mm) and its four M4 holes	They set the floor inserts	Cross-repo assumption
8	Transport cases with at least 1,250 mm inside	The wing panel is 1,205 mm with its root fittings	KWR-CAL-001 F

■ Value engineering

Value-engineering target: USD 5,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 4,293 (USD 707 under the target). Main cost drivers and savings worth trying:

- The largest lines are the Kitewright Core avionics share (USD 900, assumed), the two ColdCell packs of 5.0 Ah cells (USD 700), the four 5215-class lift motors (USD 600), the two transport cases (USD 360) and the four lift ESCs (USD 180).
- The 2026-10-03 decisions added USD 348: motors USD 120, propellers USD 40, cells USD 120, boom tubes USD 8 and the Core power leads USD 60 (moved from the Core's bill).
- Savings worth trying: soft padded bags in place of hard cases for vehicle transport (about USD 200); four-in-one ESC boards are lighter but would move power wiring into the fuselage, so they are not recommended.

■ Decisions made

Date	Decision	Decided by	Record
2026-10-03	TRL 2 review decisions D1 to D10: PX4; 2.5 m two-piece wing; one 20 in propeller set; winch on Lift; nose tractor and tail boom; two ColdCell 6S3P packs; foam, ply and printed build; Sikkim State Disaster Management Authority as first co-design candidate (not agreed); staged powered tests; conservative failsafes	Amish: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." and "Proceed with the remaining 15 scaffolds"	KWR-DDR-001
2026-10-03	Design for construction C1 to C13: tail boom, pylons, boom spacing, motors on the centre of gravity, nose tractor, joiner and wing bolts, fuselage bays, landing legs, wing conduits, payload mount, tail joint, hatch, pitot	Amish, same pre-approval	KWR-DDR-002
2026-10-03	O1 (36B), R1: 22 in propellers on 5215-class motors, booms 50 mm longer and 20 mm further out; hover margin 52.0 %	Amish: "i agree with all the 46 recommendations you provided. please proceed."	KWR-DDR-003
2026-10-03	O2 and O3 (37A), R2 and R3: 5.0 Ah cells in the same two 6S3P packs; 32.9 min and 44.3 km with O1-B	Amish, same words	KWR-DDR-003
2026-10-03	O4 (38A), R7: keep the foam, glass and lite-ply airframe and weigh it at TRL 4; 11.07 kg	Amish, same words	KWR-DDR-003
2026-10-03	Range supplies the four Kitewright Core power leads with AS150 plugs (Core decision 33B)	Amish, same words (Core decision 33B)	KWR-DDR-003
2026-10-04	Round-3 decision 10A: AS150 plugs on the packs (XT90 before); the Kitewright Core to the one family mounting envelope, the Core's drawing the reference: hung under the floor at the centre of gravity on four M4 (220 x 130 mm) in a 6 mm birch doubler, lid up through a 200 x 112 mm opening, fuselage 156 mm wide (150 inside), packs moved to the nose bay and behind the lid, rear bulkhead at 645 mm, GNSS and antennas on the hatch; Range recomputed with the 0.99 kg Core (30.6 min, 41.5 km, 48.3 %); the Kitewright interface table in docs/REVIEW.md	Amish: "For round 3, I agree with all your proposed recommendations"	KWR-DDR-004